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BCACACS 103

**First Semester B.C.A. Degree Examination, December 2025/January 2026
(SEP) (2024 – 25 Batch Onwards)**

DISCRETE MATHEMATICS FOR COMPUTER APPLICATIONS

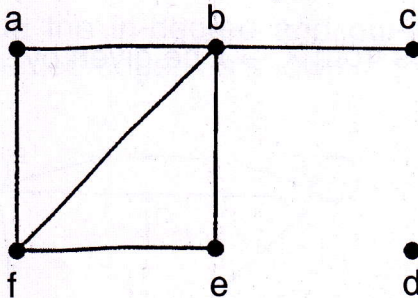
Time : 3 Hours

Max. Marks : 80

Note : Answer **any ten** questions from Part – A and **one full** question from **each** Unit in Part – B.

PART – A

1. a) Write the power set of $A = \{1, 2, 3\}$. (10×2=20)
b) Write which of these is a tautology or a fallacy.
i) $p \vee \neg p$ ii) $p \wedge \neg p$
c) Represent
i) $A \cup B$ ii) $A - B$ using Venn diagram.
d) There are 32 microcomputers in a computer center. Each microcomputer has 24 ports. How many different ports to a microcomputer in the center are there ?
e) Let $x = \{1, 5, p, \text{Jack}\}$, $Y = \{2, 5, 7, q, \text{Jill}\}$ and $f = \{\langle 1, 2 \rangle, \langle 5, 7 \rangle, \langle p, q \rangle, \langle \text{Jack}, q \rangle\}$. Find D_f and R_f .
f) How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points ?
g) What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7 ?
h) What are the quotient and remainder when 101 is divided by 11 ?
i) What is the least common multiple of $2^3 3^5 7^2$ and $2^4 3^3$?
j) How many edges are there in a graph with 10 vertices each of degree six ?
k) Find isolated and pendant node in the graph.



- l) What is binary tree ? Give an example.

P.T.O.



PART – B
Unit – I

2. a) $R = \{\langle 1, 2 \rangle, \langle 3, 4 \rangle, \langle 2, 2 \rangle\}$, $S = \{\langle 4, 2 \rangle, \langle 2, 5 \rangle, \langle 3, 1 \rangle, \langle 1, 3 \rangle\}$. Write $R \circ S$, $R \circ R$, $S \circ R$, $S \circ S$ and $R \circ (S \circ R)$.
 b) Show the following implications :
 i) $(P \wedge Q) \Rightarrow (P \rightarrow Q)$ ii) $P \Rightarrow (Q \rightarrow P)$.
 c) $A = \{\alpha, \beta\}$ and $B = \{1, 2, 3\}$. Find A^2 , B^2 , $A^2 \times B$, $A \times B$ and $B \times A$. (5+5+5)
3. a) $A = \{1, 2, 3\}$, $B = \{1, 2, 5, 7, 9\}$. Write $A - B$, $B - A$, $A + B$, $A \cup B$, $A \cap B$.
 b) $R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 2), (4, 4)\}$.
 Construct relation matrix of R and draw digraph of R .
 c) For the relation matrices
- $$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, M_S = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$
- Find $M_{R \circ S}$, $M_{\tilde{R}}$, $M_{\tilde{S}}$, $M_{\widetilde{R \circ S}}$ and show that $M_{\widetilde{R \circ S}} = M_{\tilde{S}} \circ M_{\tilde{R}}$. (5+5+5)

Unit – II

4. a) Let $X = \{2, 3, 6, 12, 24, 36\}$ and the relation $\leq$ be such that $x \leq y$ if x divides y .
 Draw the Hasse diagram of $(X, \leq)$.
 b) Show that functions $f(x) = x^3$ and $g(x) = x^{\frac{1}{3}}$ for $x \in \mathbb{R}$ are inverse of one another.
 c) A computer company receives 350 applications from computer graduates for a job planning a line of new Web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business? (5+5+5)
5. a) Let $x = \{1, 2, 3\}$ and f, g, h and s be functions from $X \rightarrow x$ be given by
 $f = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 1 \rangle\}$
 $g = \{\langle 1, 2 \rangle, \langle 2, 1 \rangle, \langle 3, 3 \rangle\}$
 $h = \{\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 1 \rangle\}$
 $s = \{\langle 1, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 3 \rangle\}$
 Find $f \circ g$, $g \circ f$, $f \circ g \circ h$, $s \circ g$, $s \circ s$.





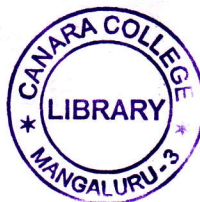
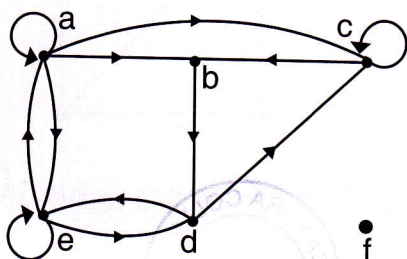
- b) A multiple-choice test contains 10 questions. There are four possible answers for each question.
- i) In how many ways can a student answer the questions on the test if the student answers every question ?
 - ii) In how many ways can a student answer the questions on the test if the student can leave answers blank ?
- c) Define Surjective, Injective and Bijective functions with an example. (5+5+5)

Unit – III

6. a) Find the prime factorization of 7007.
- b) What is the variance of the random variable $X((i, j)) = 2i$, where i is the number appearing on the first die and j is the number appearing on the second die, when two fair dice are rolled ?
- c) Use mathematical induction to prove the inequality $n < 2n$ for all positive integers n . (5+5+5)
7. a) Find the greatest common divisor of 414 and 662 using the Euclidean algorithm.
- b) Suppose that we train a Bayesian spam filter on a set of 2000 spam messages and 1000 messages that are not spam. The word “stock” appears in 400 spam messages and 60 messages that are not spam, and the word “undervalued” appears in 200 spam messages and 25 messages that are not spam. Estimate the probability that an incoming message containing both the words “stock” and “undervalued” in spam, assuming that we have no prior knowledge about whether it is spam. Will we reject such messages as spam when we set the threshold at 0.9 ?
- c) What is the conditional probability that a family with two children has two boys, given they have at least one boy ? Assume that each of the possibilities BB, BG, GB and GG is equally likely, where B represents a BOY and G represents a GIRL. (Note that BG represents a family with an older Boy and younger Girl while GB represents a family with older Girl and younger Boy). (5+5+5)

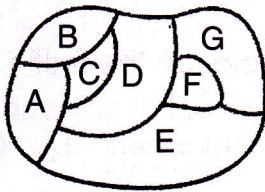
Unit – IV

8. a) Find the in-degree and out-degree of each vertex in the graph G with directed edges as shown in the figure.



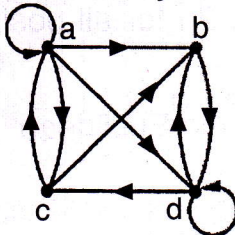


- b) Construct the dual graph for the map. Then find the number of colours needed to colour the map so that no two adjacent regions have the same colour.

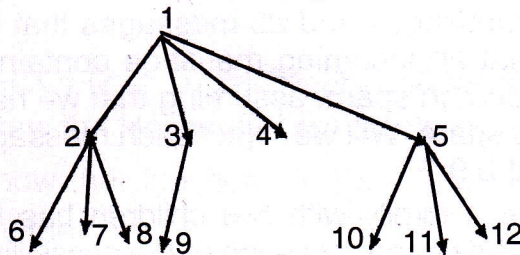


- c) Give the state diagram of the finite state automata of a device that compares two binary numbers to determine whether they are equal, or which of the two is larger. Consider the input alphabet as $\{00, 01, 10, 11\}$ and output alphabet as $\{EQUAL, LARGER, SMALLER\}$. (5+5+5)

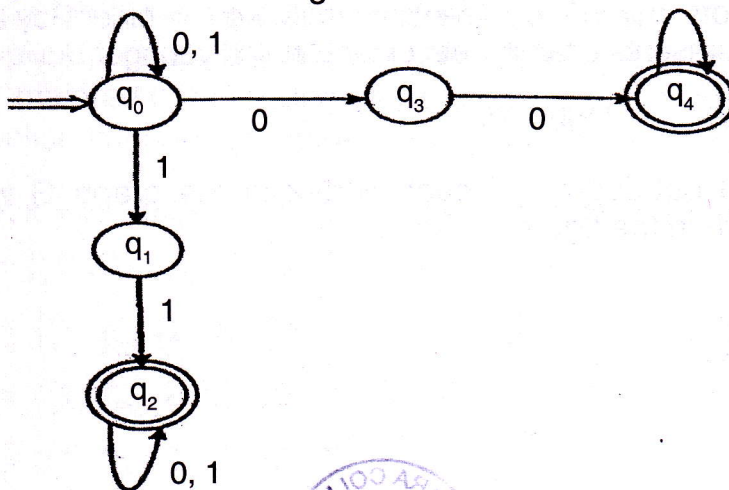
9. a) Use an adjacency list to represent the given graph.



- b) Convert the following tree into a binary tree.



- c) Convert the following NFA to DFA.



(5+5+5)

